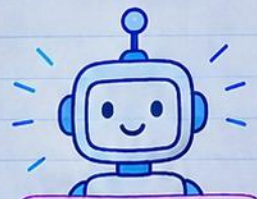


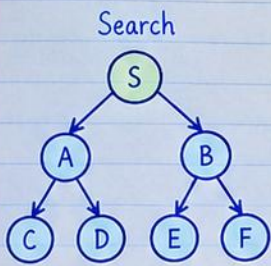
Artificial Intelligence: A Modern Approach

Second Edition

Stuart Russell & Peter Norvig



"Agents perceive, reason, act, and learn in complex environments."



Explore states to find a goal!

Handwritten Study Notes

Batch 1: Cover + Contents + Chapter 1 + Chapter 2

27 Chapters

Compact Exam-Friendly Review

1. Problem-Solving and Search

- Formulate problems as state spaces.
- Uninformed and informed search.
- Optimal solutions and heuristics.
- Applications: puzzles, games, routing.

2. Knowledge and Reasoning

- Logical representation of knowledge.
- Propositional and first-order logic.
- Inference, resolution and planning.
- Handling incomplete and uncertain knowledge.

$$\begin{array}{l} P \Rightarrow Q \\ P \\ \hline \therefore Q \end{array}$$

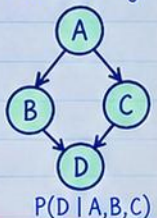
3. Planning and Decision Making

- Plan sequences of actions.
- Classical and probabilistic planning.
- Markov decision processes (MDPs).
- Sequential decision making.

Start → ○ → ○ → Goal

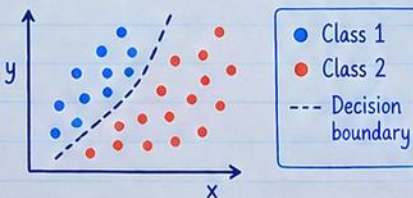
4. Uncertainty and Probabilistic Reasoning

- Represent and reason under uncertainty.
- Bayesian networks.
- Probabilistic inference.
- Hidden Markov models.
- Decision making under uncertainty.



5. Learning from Experience

- Learn from data.
- Supervised and unsupervised learning.
- Reinforcement learning.
- Generalization and model evaluation.



6. Language, Perception and Robotics

- Natural language processing.
- Perception from noisy, real-world data.
- Computer vision and pattern recognition.
- Robotics and acting in physical environments.



Language understanding



Perception (vision)



Robotics and action

Intelligent Agents for a Complex World



Perceive → Reason → Act → Learn

Why These Notes?

- Clear and concise summaries.
- Diagrams and examples for quick revision.
- Focus on key ideas, algorithms and intuition.
- Useful for exams, interviews and self-study.

Who is this for?

- Students taking AI courses.
- Readers of AIMA (Second Edition).
- Anyone who wants a compact, handwritten revision guide.

What's Inside (Batch 1)?

- Cover page (this page)
- Table of contents (next page)
- Chapter 1: Introduction to AI
- Chapter 2: Intelligent Agents

"Understand concepts, compare algorithms, and revise smarter."

Part I : Artificial Intelligence (Ch 1 - 2)

Introduces AI, its goals, and the agent perspective.

Chapter	Title
1	Introduction
2	Intelligent Agents

Part II : Problem-solving (Ch 3 - 6)

Studies general problem-solving methods, including search, constraint satisfaction and game playing.

Chapter	Title
3	Solving Problems by Searching
4	Informed Search and Exploration
5	Constraint Satisfaction Problems
6	Adversarial Search

Part III : Knowledge and Reasoning (Ch 7 - 10)

Describes logical agents, formal logic, inference and knowledge representation.

Chapter	Title
7	Logical Agents
8	First-Order Logic
9	Inference in First-Order Logic
10	Knowledge Representation

Part IV : Planning (Ch 11 - 12)

Studies planning methods and how to act in complex, real-world environments.

Chapter	Title
11	Planning
12	Planning and Acting in the Real World

Appendices

Provide additional background material.

Appendix	Title
A	Mathematical Background
B	Notes on Languages and Algorithms

Part V : Uncertain Knowledge and Reasoning (Ch 13 - 17)

Handles reasoning and decision making under uncertainty using probabilistic models.

Chapter	Title
13	Uncertainty
14	Probabilistic Reasoning
15	Probabilistic Reasoning over Time
16	Making Simple Decisions
17	Making Complex Decisions

Part VI : Learning (Ch 18 - 21)

Covers learning from data, knowledge in learning, and statistical and reinforcement learning methods.

Chapter	Title
18	Learning from Observations
19	Knowledge in Learning
20	Statistical Learning Methods
21	Reinforcement Learning

Part VII : Communicating, Perceiving, and Acting (Ch 22 - 25)

Covers natural language communication, perception, and robotic action.

Chapter	Title
22	Communication
23	Probabilistic Language Processing
24	Perception
25	Robotics

Part VIII : Conclusions (Ch 26 - 27)

Discusses the philosophical foundations of AI and its future directions.

Chapter	Title
26	Philosophical Foundations
27	AI: Present and Future

* Current batch covers:

- Chapter 1 : Introduction
- Chapter 2 : Intelligent Agents



Planned format:

4 handwritten pages per chapter.

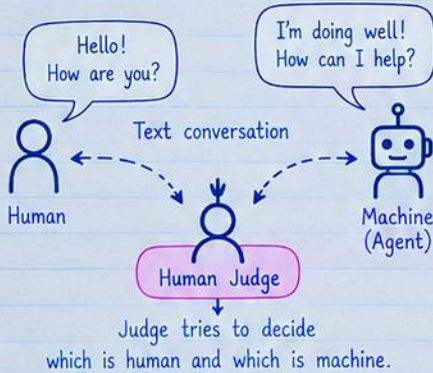
1. What is AI?

- AI is the study of intelligent agents and the design of machines that perceive, reason, learn, and act.
- Goal: build systems that can perform tasks that normally require human intelligence.
- An intelligent agent perceives its environment through sensors and acts through actuators to achieve goals.

Key Idea: AI studies how to create agents that act rationally in complex environments.

4. Acting humanly: Turing Test

- A machine exhibits intelligent behavior if a human judge cannot distinguish its responses from those of a human.
- The test is based on natural language conversation.



Total Turing Test: In addition to language, the agent must also have perception (e.g., vision) and robotics (e.g., movement in the physical world).

2. Why AI matters

- Automation:** perform repetitive or dangerous tasks (e.g., manufacturing).
- Decision support:** assist human experts with better information (e.g., medical diagnosis).
- Pattern recognition:** find structure in large amounts of data (e.g., images, speech).
- Robotics:** interact with the physical world.
- Language:** understand and generate human language.
- Everyday systems:** search engines, maps, recommendations, smart assistants.

5. Thinking humanly

- Focuses on modeling human thought processes.
- Uses findings from psychology, neuroscience, and cognitive science.
- Builds computational models of how humans perceive, remember, reason, and learn.

Example: A model of human memory or problem solving can inspire algorithms for AI systems.

7. Acting rationally

- An agent should choose the action that maximizes its expected success (or utility), based on its percepts and knowledge.
- Rational behavior is what we ideally want, even if it is not human-like.
- Takes into account goals, uncertainty, and the consequences of actions.

Key Idea: A rational agent does the best thing to achieve its goals, given what it knows and what it perceives.

3. The four classic views of AI

- Thinking humanly**
Model how humans think.
- Acting humanly**
Behave like humans.
- Thinking rationally**
Use logical reasoning.
- Acting rationally**
Choose actions to achieve goals as well as possible.

6. Thinking rationally

- Based on the laws of thought.
- Uses formal logic to represent knowledge and reason.
- A system should derive correct conclusions from given information using valid inference rules.

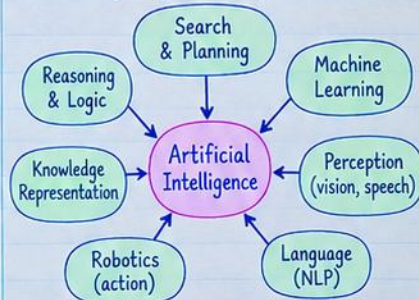
Example: If we know
All birds fly.
Tweety is a bird.
Therefore, Tweety flies.

Uses:

- Propositional logic
- First-order logic
- Knowledge representation
- Logical inference

8. AI as a broad field

- AI brings together many subfields:



These areas work together to build intelligent agents that operate in the real world.

9. Comparison of the four approaches

Approach	Main Idea	Focus	Methods / Examples	Strengths	Limitations
Thinking humanly	Model human thought processes	How humans think	Cognitive models, neuroscience	Matches human behavior	Human cognition is not fully understood
Acting humanly	Behave like humans	Human-like behavior	Turing Test, NLP, perception, robotics	Intuitive and natural interface	Imitation does not guarantee true intelligence
Thinking rationally	Use logical reasoning	Correct reasoning	Formal logic, inference rules	Clear, well-defined foundation	May not handle uncertainty or real-world data easily
Acting rationally	Maximize expected success	Goal-directed action	Search, planning, probabilistic reasoning, reinforcement learning	Most general and widely applicable	Requires defining goals and utility; can be complex in real environments

10. Key takeaway

- AI is the study of intelligent agents that perceive, reason, learn, and act.
- AI matters because it enables automation, better decisions, recognition, robotics, language, and many everyday systems.
- There are four classic views: thinking humanly, acting humanly, thinking rationally, acting rationally.
- Each view offers a useful perspective, and together they provide a broad foundation for the field.

“AI is about building agents that perceive, reason, learn, and act to achieve goals.”

— AIMA (Ch.1)

1. Foundations of AI

- AI studies how to build intelligent agents that perceive, reason, learn and act.
- Draws ideas from many fields (science, engineering, mathematics, and humanities).
- Seeks general principles of intelligence and practical methods to solve real-world problems.

Key Idea: AI is built on a rich foundation of ideas from many disciplines.

Examples: learning, planning, language understanding, perception, decision making.

2. Philosophy

- Logic: rules of correct reasoning.
- Mind: nature of thought, consciousness and intelligence.
- Knowledge: what it means to know something; representation of knowledge.
- Rationality: acting according to reasons to achieve goals.

Questions for AI

- What is intelligence?
- How should an agent reason?
- What does it mean to be rational?
- Can a machine have a mind?

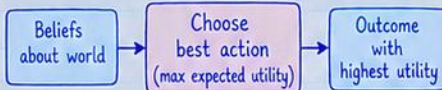
3. Mathematics

- Formal logic: precise representation and inference.
- Probability: reasoning under uncertainty.
- Statistics: learning from data.
- Algorithms: step-by-step procedures for problem solving.
- Computation: what can be computed and how efficiently.
- Optimization: finding the best solution (subject to constraints).

Provides the language and tools for building and analyzing AI methods.

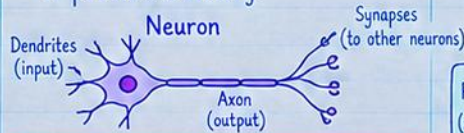
4. Economics / Decision Theory

- Utilities: numerical values for preferences.
- Preferences: ordering of alternatives.
- Rational choice: choose action that maximizes expected utility.
- Game theory: strategic interaction among multiple agents.



5. Neuroscience

- Studies how brains work and generate intelligent behavior.
- Provides inspiration for learning, perception and representation.
- Neural mechanisms: neurons, synapses, networks, plasticity.
- Helps explain how information is processed in parallel and robustly.



Brains inspire AI, but AI is not just a copy of the brain.

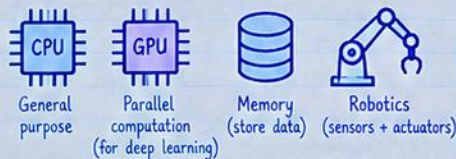
6. Psychology / Cognitive Science

- Studies human cognition and behavior.
- Topics: perception, attention, memory, problem solving, learning, reasoning, language, and decision making.
- Provides models of how people think and learn.
- Helps us design AI systems that are more natural, robust and human-centered.



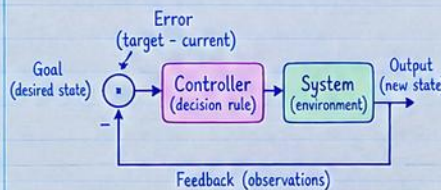
7. Computer Engineering

- Provides the hardware and software platforms for AI systems.
- Topics: computer architecture, processors (GPU/TPU), memory, sensors, networks.
- Enables efficient computation at scale.
- Important for real-time and embedded AI.



8. Control Theory & Cybernetics

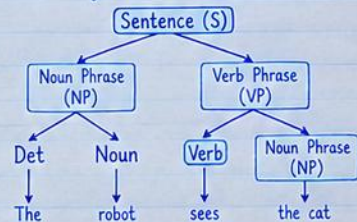
- Studies how to design systems that achieve desired behavior over time.
- Key ideas: feedback, stability, robustness, and goal-directed behavior.
- Used in robotics, autonomous systems, and sequential decision making.



Feedback helps correct errors and keep the system on track.

9. Linguistics

- Studies the structure and meaning of human language.
- Provides theories of grammar, syntax, semantics and pragmatics.
- Helps build language understanding and generation systems.
- Inspires representations for text, knowledge and communication.



10. Summary: Disciplines and Their Contributions to AI

Discipline	Key Contribution to AI
Philosophy	Logic, mind, knowledge representation, rationality, ethics.
Mathematics	Formal logic, probability, statistics, algorithms, computation, optimization.
Economics / Decision theory	Utilities, preferences, rational choice, game theory (multi-agent settings).
Neuroscience	Understanding brains; inspiration for learning and perception.
Psychology / Cognitive science	Models of human cognition, memory, problem solving, perception.
Computer engineering	Hardware, architectures, sensors, efficient computation, robotics.
Control theory & cybernetics	Feedback, stability, goal-directed behavior, autonomous control.
Linguistics	Language structure, meaning, and representation for natural language processing.

Big Idea: AI is interdisciplinary.

- AI brings together ideas from many fields.
- Each discipline provides a different piece of the puzzle.
- Together, they help us build intelligent systems that can perceive, reason, learn and act in complex, real-world environments.
- Understanding these foundations helps us design better AI methods and recognize their strengths and limits.

1. A Brief History of AI

Early optimism (1940s-1950s)

- Idea that human intelligence can be replicated.
- Early programs for playing games and theorem proving.

Symbolic AI (1950s-1970s)

- Knowledge represented with symbols and logic.
- Focus on search, reasoning and planning.

Expert systems (1980s)

- Domain specific knowledge bases (e.g. medical diagnosis).
- Success in narrow domains, high expectations.

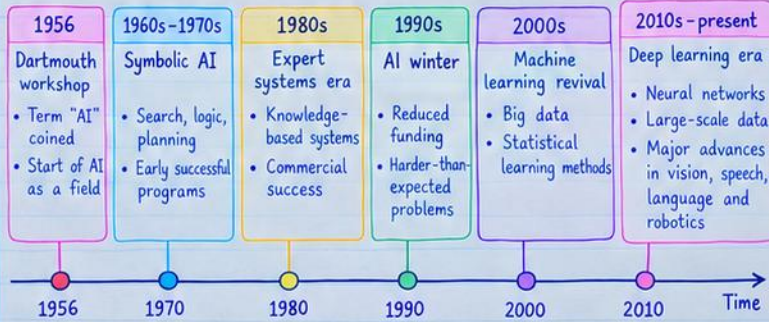
AI winters (late 1980s-1990s)

- Limited progress, overhyped claims.
- Reduced funding and interest.

Modern data-driven AI (2000s-present)

- Machine learning with large datasets.
- Deep learning and representation learning.
- Major advances in perception, language and robotics.

2. Timeline of Key Milestones



3. What Makes AI Difficult?

- Huge search spaces**
 - Many possible states and actions
- Uncertainty**
 - Incomplete and noisy information
- Incomplete knowledge**
 - The real world is complex and not fully known
- Perception**
 - Processing unstructured data (images, sound, text)
- Learning from data**
 - Need to generalize and handle variation



Real-world problems have enormous search spaces and uncertain, incomplete information.

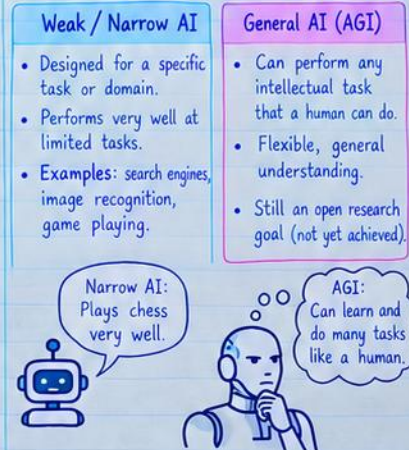
4. Main Capabilities of AI

- Problem solving** - finding sequences of actions.
- Knowledge representation** - storing knowledge about the world.
- Reasoning** - deriving new information.
- Learning** - improving from experience.
- Communication** - understanding and generating human language.
- Perception** - interpreting sensory input (vision, speech, etc.).
- Acting** - taking actions in the real world.

5. Common AI Tasks



6. Weak/Narrow AI vs General AI



7. Performance and Evaluation

- Define a performance measure for each task.
- Common measures: accuracy, success rate, speed (time), cost, and resource usage.
- Evaluate on test data or real-world environments.
- Consider robustness and generalization.
- Sometimes compare to human-level performance.

Example: Image classification

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of test examples}}$$

Confusion Matrix (example)

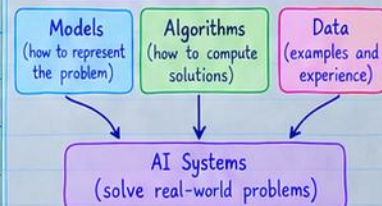
	Predicted cat	Predicted dog
Actual cat	45	5
Actual dog	7	43

8. Symbolic Methods vs Data-Driven Methods

	Symbolic Methods (knowledge-based)	Data-Driven Methods (machine learning)
Representation	Symbols, logic, rules	Numeric data, feature vectors
Knowledge source	Human knowledge (engineered)	Learned from data
Reasoning	Logical inference, search	Statistical inference, pattern recognition
Strengths	Explainable, uses explicit knowledge	Handles large, complex data; good at perception
Weaknesses	Hard to scale, brittle in real world	Needs large data, less interpretable
Examples	Logic, rule-based systems, search	Neural networks, probabilistic models

9. AI is Not Magic

AI is not magic; it combines models, algorithms, and data to solve problems.



10. Summary

- AI has a long history with periods of excitement and setbacks.
- The goal is to build intelligent agents that can perceive, reason, learn, and act.
- AI is difficult because of large search spaces, uncertainty, incomplete knowledge, and real-world complexity.
- AI includes a wide range of tasks from game playing to vision, language, and robotics.
- There are different approaches, including symbolic and data-driven methods.
- Success is measured using appropriate performance metrics.
- General AI remains a major long-term goal.

1. Intelligent Agent Preview

- An intelligent agent perceives its environment through sensors and acts on it through actuators to achieve goals.
- The agent selects actions that are rational (i.e., expected to maximise performance).
- Chapter 2 will formally define agents and different types of agent programs.

Key Idea: AI studies how to design agents that perceive, reason, learn and act in the real world.

2. State of the Art

Area	Examples / Current Capabilities
Search & planning	Solves large puzzles, route planning, scheduling, resource allocation.
Games	Superhuman play (e.g., chess, Go, poker, video games).
Language	Conversational agents, translation, summarization, code generation.
Vision	Object recognition, image generation, scene understanding.
Medicine	Medical image analysis, diagnosis support, drug discovery.
Autonomous systems	Self-driving cars, drones, robots in complex environments.

Note: Rapid progress is driven by better algorithms, more data and greater computing power.

3. Benefits and Concerns

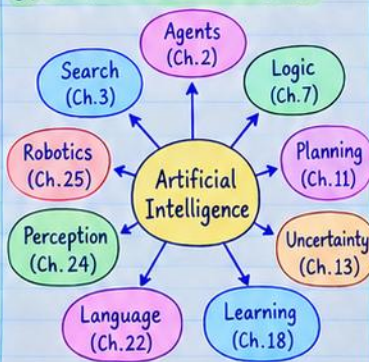
Potential Benefits	Key Concerns
Higher productivity and automation	Job displacement and economic impact
Scientific discovery (e.g., drugs, materials)	Bias and unfair outcomes
Better decision support (e.g., medicine, climate)	Safety and robustness in real-world use
Personalized services (education, healthcare)	Privacy and data misuse
Solving complex global problems (e.g., environment)	Lack of transparency and explainability

Key Point: We want to maximise benefits while responsibly addressing the risks.

4. AI as Science vs AI as Engineering

AI as a Science	AI as an Engineering Discipline
- Understand intelligence and behaviour. - Study principles of reasoning, learning, perception and action. - Build models of natural and artificial intelligence. - Use insights from psychology, neuroscience, mathematics and computer science.	- Design and build useful intelligent systems. - Turn theories into algorithms and working applications. - Handle real-world constraints (e.g., time, memory, reliability). - Deploy systems that are robust, scalable and beneficial. - Involves system design, evaluation and iteration.

5. Core Themes of the Book



These themes work together in many real-world applications, such as autonomous robots or assistants.

6. Important Terms to Remember

Term	Definition (short)
Artificial Intelligence (AI)	The study of intelligent agents that perceive, reason, learn and act to achieve goals.
Rationality	Choosing the action expected to maximise performance.
Turing Test	A test for machine intelligence based on indistinguishable conversation.
Agent	An entity that perceives through sensors and acts through actuators.
Environment	Everything the agent interacts with, outside the agent.
Learning	Improving performance from experience.
Reasoning	Drawing conclusions from knowledge to make decisions.

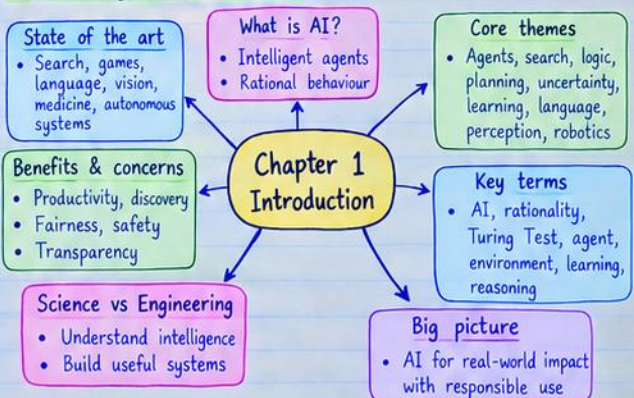
7. Likely Exam Questions

- Define an intelligent agent. What are its components?
- Explain what is meant by a rational agent.
- How does AI as a science differ from AI as an engineering discipline?
- List and briefly explain the core themes covered in this book.
- Give examples of current AI applications in different domains.
- Discuss the potential benefits and concerns of AI.
- What is the Turing Test? What does it measure?
- Which AI topics would you use for a given real-world problem (e.g., self-driving car or medical diagnosis)? Give reasons.

9. Key Takeaways

- AI aims to build intelligent agents that act rationally.
- AI is both a science (understanding intelligence) and an engineering discipline (building useful systems).
- Progress is visible in search, games, language, vision, medicine and autonomous systems.
- AI brings major benefits but also raises important concerns (fairness, safety, transparency).
- The book covers a coherent set of core themes from agents to robotics.
- Key terms such as rationality, Turing Test, agent, environment, learning and reasoning form the foundation for later chapters.
- Many real-world problems require a combination of these AI techniques.
- Responsible and human-centered AI is essential for long-term societal benefit.

8. Memory Aid / Mind Map



“Intelligence is not just about knowing, but about choosing the right action to achieve a goal in the real world.”
— AIMA (Ch.1)

1. What is an agent?

- An agent is anything that perceives through sensors and acts through actuators.
- An agent receives information from the environment, decides what to do, and takes actions to affect the environment.

Key Idea: An agent is an entity that perceives and acts.

2. Agent and environment

- Environment:** everything outside the agent.
- Percepts:** information received from the environment through sensors.
- Actions:** things the agent can do through actuators.
- Percept sequence:** the complete history of all percepts an agent has received so far.

3. Agent function

- An agent function maps each possible percept sequence to an action.
- It specifies what the agent should do for every possible history of percepts.

Agent function: $f : P^* \rightarrow A$

P^* : set of all possible percept sequences
 A : set of possible actions

Example: For a vacuum cleaner agent, if the percept sequence says "current square is dirty", the agent function maps to the action "Suck".

4. Agent program

- An agent program is an implementation of the agent function on a physical or virtual architecture.
- It takes the current percept (or percept history) as input and returns an action.
- The same agent function can be implemented by many different programs.

Agent program: runs on an architecture to implement the agent function.

6. Rational agent

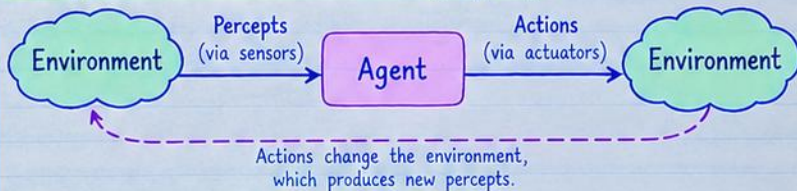
- A rational agent chooses actions that maximize the expected value of a performance measure, given the available information (the percept sequence).
- It does the best it can, even when the environment is uncertain, partially observable, or changing.

Performance measure: a way to evaluate how well the agent is doing (e.g., total reward, accuracy, speed, safety).

5. Sensors and actuators

Type of agent	Sensors (percepts)	Actuators (actions)
Human	Eyes (vision), ears (sound), skin (touch), nose (smell), tongue (taste)	Hands and fingers (move, grasp), legs (walk), speech (talk)
Robot	Cameras, microphones, range finders (distance), touch sensors, GPS	Motors (move wheels/arms), grippers, speakers, display lights
Software agent	Keyboard input, mouse clicks, files, network data, API calls	Display output, send messages, modify files, make API requests, recommend items

7. Agent-environment interaction (diagram)



8. Agent function vs. agent program

Agent function (what to do)	Agent program (how to do it)
- Abstract mathematical mapping from percept sequences to actions. - Specifies the ideal behavior. - Independent of the implementation. - Many different programs can implement the same agent function.	- Concrete implementation on an architecture (computer, robot, etc.). - Takes percepts as input and outputs an action. - Considers computational resources (memory, time). - One specific program implements the agent function.

9. Examples of intelligent agents

Vacuum cleaner agent



- Percepts: dirt sensor, location sensor
- Actions: suck, move left, move right
- Goal: keep the floor clean

Self-driving car



- Percepts: cameras, lidar, radar, GPS
- Actions: steer, accelerate, brake
- Goal: reach destination safely and efficiently

Chess program



- Percepts: board positions
- Actions: make a move
- Goal: win the game

Recommendation system



- Percepts: user history, clicks, ratings
- Actions: recommend items
- Goal: suggest items the user is likely to like (maximize user satisfaction)

10. Summary (Chapter 2 Page 1)

- An agent perceives through sensors and acts through actuators.
- The agent function maps percept sequences to actions; the agent program implements it on an architecture.
- A rational agent chooses actions that maximize expected performance given the available information.
- Intelligent agents exist in many forms, from simple vacuum cleaners to complex systems like self-driving cars and recommender systems.

1. What is rationality?

- A rational agent does the right thing based on its percepts (evidence) and its goals.
- It chooses the action expected to maximize the value of the performance measure, given what it knows.
- Rationality is about acting to achieve goals, not about thinking like a human.

Key Idea: Rationality means choosing actions that are expected to be best, given the evidence and available information.

2. Performance measure

- Defines what counts as success for the agent.
- Specified externally by the designer.
- Used to evaluate how well the agent is doing.
- Different performance measures can lead to very different agent behaviors.

Examples:

- Taxi: reach destination safely and quickly.
- Medical diagnosis: correct and timely diagnosis.
- Vacuum cleaner: keep rooms clean with low cost.

3. Factors affecting rationality

The choice of action depends on:

- Performance measure** – what is considered success.
- Prior knowledge** – what the agent already knows about the world.
- Percept history** – what the agent has observed so far.
- Available actions** – what actions the agent can perform.

4. Rationality vs. omniscience

- A rational agent is not all-knowing.
- It acts rationally based on the information available to it.
- It may make mistakes if the information is incomplete or uncertain.
- Omniscience (knowing everything) is not required and is often impossible in real-world environments.

Example: A taxi agent does not know future traffic, but it can still act rationally using current percepts and knowledge.

5. Learning and autonomy

- Good agents can improve their performance from experience.
- Learning allows the agent to update its knowledge about the environment.
- More experience often leads to better decisions.
- Autonomy means the agent can operate for extended periods without human intervention, using its own experience.

Key Point: Learning helps an agent become more rational over time, especially in unknown or changing environments.

6. PEAS description

A task environment is specified using PEAS:

- P** Performance measure : criteria for success
- E** Environment : what the agent interacts with
- A** Actuators : actions the agent can perform
- S** Sensors : what the agent can perceive

PEAS helps clarify the task and guides the design of the agent, its sensors and actuators, and the necessary knowledge and algorithms.

7. PEAS examples for different intelligent agents

Agent Type	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Autonomous Taxi	- Reach destination safely - Minimize time and cost - Obey traffic rules - Maximize passenger comfort	- City streets and highways - Other vehicles, pedestrians - Traffic signals, road conditions - Passengers, weather	- Steering - Acceleration / braking - Turn signals - Horn, display messages	- Cameras - Lidar / radar - GPS, maps - Speedometer, inertial sensors
Medical Diagnosis System	- Correct diagnosis - Timely treatment recommendation - Minimize test cost and risk - Improve patient health	- Patients - Diseases and medical knowledge - Test results, medical history - Dealing with uncertainty	- Ask for tests - Recommend treatments - Provide explanations to doctor - Display advice	- Patient symptoms (input) - Test results (lab, imaging, etc.) - Medical records - Input from doctors
Robotic Vacuum Cleaner	- Keep rooms clean - Minimize time and energy - Avoid collisions - Minimize wear and tear	- Indoor home/office environment - Furniture, walls, people, pets - Dirt, different floor types	- Move (wheels) - Turn - Suction / cleaning brush - Dock for charging	- Bump sensors - Cameras or proximity sensors - Dirt sensors - Odometry (wheel encoders)

Specify the task environment before designing the agent.

A clear PEAS description helps us understand what the agent should achieve and what information and actions are available.

8. Common mistakes in defining performance measures

- Rewarding the wrong thing – leads to unintended behavior.
Example: a taxi that goes in circles to maximize distance traveled.
- Missing important aspects – safety, cost, fairness, or long-term effects may be ignored.
- Measures that are easy to game – agent finds loopholes instead of achieving the real goal.
- Not considering side effects – agent may optimize one measure but cause harm in other ways.
- Too narrow or too broad – measure may not reflect the true objective.

9. Summary / Key Points

- A rational agent chooses actions that are expected to maximize the performance measure, given its percepts and knowledge.
- The performance measure defines success from the outside.
- Rationality depends on the performance measure, prior knowledge, percept history, and available actions.
- Rationality does not require omniscience.
- Learning and autonomy help agents improve performance over time.
- PEAS provides a clear way to specify the task environment.
- A good performance measure is essential for the desired behavior.
- Always specify and understand the task environment before designing the agent.

“A rational agent makes the best possible decision, given what it knows, to achieve its goals.” — AIMA (Ch.2)

1. Fully Observable vs Partially Observable

- Fully observable: agent can see the complete state of the environment at each time step.
- Partially observable: agent has incomplete or noisy information about the state.

Example:

Fully observable



Agent sees entire map.

Partially observable



Agent sees only nearby cells (others are unknown).

2. Deterministic vs Stochastic

- Deterministic: next state is completely determined by current state and action.
- Stochastic: next state is described by a probability distribution (uncertain outcomes).

Example:

Deterministic



Same action always leads to B.

Stochastic



Action can lead to different results.

3. Episodic vs Sequential

- Episodic: each decision is independent of previous decisions.
- Sequential: current decisions affect future decisions.

Example:

Episodic (image classification)



Classify each image independently.

Sequential (game playing)



Current move affects future moves.

4. Static vs Dynamic

- Static: environment does not change while the agent is deliberating.
- Dynamic: environment can change during deliberation (time matters).

Example:

Static (puzzle)



State remains the same while thinking.

Dynamic (driving)



Other cars and pedestrians can move while agent decides.

5. Discrete vs Continuous

- Discrete: finite, countable set of states, actions or time steps.
- Continuous: states, actions or time vary over a continuous range of values.

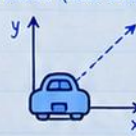
Example:

Discrete (grid world)



Moves to one of four neighboring cells.

Continuous (robot control)



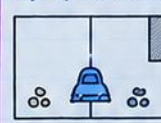
Can move in any direction and any distance.

6. Single-Agent vs Multiagent

- Single-agent: only one agent acts in the environment.
- Multiagent: multiple agents (cooperative or competitive) interact.

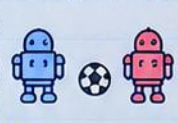
Example:

Single-agent (vacuum world)



Only one agent cleans the environment.

Multiagent (soccer)



Multiple agents with possibly competing goals.

7. Known vs Unknown Environment

- Known: agent knows the rules of the environment (transition model, percepts, rewards).
- Unknown: agent must learn or explore to understand how the environment works.

Example:

Known (chess)



Rules are known in advance.

Unknown (exploration)



Agent must explore to learn the rules.

8. Summary of Environment Dimensions

Dimension	First Type	Second Type	Simple Example
Observability	Fully observable	Partially observable	Chess (full) vs. Taxi driving (partial)
Determinism	Deterministic	Stochastic	8-puzzle (det.) vs. Taxi driving (stoch.)
Task structure	Episodic	Sequential	Image classification (episodic) vs. Chess (sequential)
Dynamics	Static	Dynamic	8-puzzle (static) vs. Taxi driving (dynamic)
State/Action space	Discrete	Continuous	Chess (discrete) vs. Robot control (continuous)
Number of agents	Single-agent	Multiagent	Vacuum world (single) vs. Soccer (multi)
Knowledge	Known	Unknown	Chess (known) vs. Exploration (unknown)

9. Example Task Classifications

Task	Observable?	Deterministic?	Episodic?	Static?	Discrete?	Multiagent?	Known?
Chess	Fully	Deterministic	Sequential	Static	Discrete	Multiagent (competitive)	Known
Taxi driving	Partially	Stochastic	Sequential	Dynamic	Continuous	Multiagent (mixed)	Unknown
Medical diagnosis	Partially	Stochastic	Sequential	Dynamic	Discrete (or continuous)	Single-agent	Partially unknown
Vacuum world	Fully	Deterministic	Sequential	Static	Discrete	Single-agent	Known

10. Key Takeaways

- Environment properties affect the design of the agent (sensors, actuators, reasoning and learning methods).
- No single agent design works well for all environments.
- Consider each dimension to choose the right approach (e.g., model-based, learning, multiagent reasoning).
- More difficult environments (partially observable, stochastic, dynamic, continuous, multiagent, unknown) require more advanced agent designs.

“The right agent design depends on the properties of the task environment.”

– AIMA (Ch.2)

1. Simple Reflex Agents

- Act only on the current percept using condition-action rules.
- Do not maintain any internal state.
- Suitable for fully observable, static environments.

Key Idea: A set of rules of the form "IF percept THEN action".

Example: Thermostat: if temperature > T, turn AC on.

2. Model-based Reflex Agents

- Maintain an internal state (model) of the world.
- Update state based on current percepts and a model of how the world evolves.
- Can handle partially observable environments.

Key Idea: Use internal state to decide action, not just the current percept.

Example: Robot vacuum that remembers which rooms are clean/dirty.

3. Goal-based Agents

- Have explicit goals describing desired situations.
- Choose actions that lead to goal states (typically using search or planning).
- More flexible than reflex agents.

Key Idea: Select actions that achieve the goal state.

Example: Navigation agent that plans a route to a destination.

4. Utility-based Agents

- Use a utility function to evaluate desirable outcomes.
- Choose actions that maximize expected utility.
- Can handle trade-offs and multiple goals.

Key Idea: Compare possible outcomes using a utility (preference) measure and choose the best one.

Example: Self-driving car balancing safety, speed, and comfort.

5. Learning Agents

- Improve their performance with experience.
- Can learn from interaction with the environment.
- Can be combined with any of the other agent types.

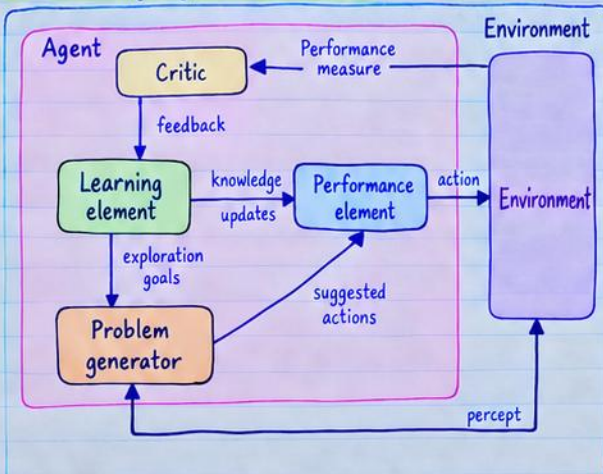
Key Idea: Use experience to get better at selecting actions.

Example: Game playing agent that learns better strategies over time.

6. Components of a Learning Agent

- Performance element**
 - Selects actions using current knowledge (e.g., a reflex, goal-based or utility-based agent).
- Learning element**
 - Improves the performance element based on experience.
- Critic**
 - Evaluates how well the agent is doing (using a performance measure).
- Problem generator**
 - Suggests exploratory actions to gain informative experience (exploration).

7. Learning Agent Architecture



8. Comparison of Agent Types

Agent Type	Uses Internal State?	Uses Goals?	Uses Utility?	Learns?	Strengths	Limitations
Simple Reflex	No	No	No	No	Very simple and fast	Works only in fully observable, static worlds
Model-based Reflex	Yes	No	No	No	Handles partial observability	Still reactive; no explicit goals
Goal-based	Yes	Yes	No	No	Flexible; can plan to achieve goals	Can be computationally expensive
Utility-based	Yes	Yes	Yes	No	Handles trade-offs; better decisions	Requires a good utility function
Learning	Yes	Maybe*	Maybe*	Yes	Improves with experience; adapts to change	Needs sufficient experience; can be complex

* Learning agents can incorporate goals and/or utility, so "Maybe" means it can be combined with those ideas.

9. When to Use Which Agent?

- Simple reflex:** Use in simple, well-defined, fully observable, and static environments.
- Model-based reflex:** Use when the environment is partially observable and a model helps.
- Goal-based:** Use when you have clear goals and need to plan or search for solutions.
- Utility-based:** Use when there are multiple goals or trade-offs, and you can define a utility function.
- Learning:** Use when the environment is complex, changing or initially unknown, and improvement over time is important.

10. Chapter 2 Recap: Key Takeaways

- Agents perceive the environment and act to achieve objectives.
- Simple reflex agents use condition-action rules.
- Model-based reflex agents maintain an internal state.
- Goal-based agents choose actions to achieve goals.
- Utility-based agents use a utility function to select the best outcomes.
- Learning agents improve from experience and consist of a performance element, learning element, critic, and problem generator.

Exam Tip:

- Be clear about the main idea and key differences between agent types.
- Know the components and diagram of the learning agent.
- Be able to explain when each agent type is appropriate and their strengths/limitations.
- Diagrams and comparison tables are commonly asked in exams!